LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future

Petroleum Refining for Non-Technical Persons

Understanding Refinery Processes, Product Yields and Margins in Plain Language

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Petroleum Refining for Non-Technical Persons is a 3-day introductory course that explains the basics of refining processes, operations and configurations in clear, non-technical language. It gives participants practical knowledge of how crude oil is processed in refineries, how different processing units and configurations affect product yields and margins, and why certain feedstocks are chosen over others.

The course demonstrates the role of key process units and operations in maximising refining margins under different crude and condensate feeds, and shows how refined products, grades, specifications, supply, trading and marine freight fit into the broader downstream business. Using real-life examples from refining and marketing organisations, the programme bridges theory and practice so non-technical professionals can understand how refineries operate, how products are made and blended, and how refining economics link to market prices and benchmarks.

LOCATION

Rio de Janeiro

London

Singapore

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COURSE OBJECTIVES

By the end of the course, participants will be able to:

- Explain the fundamentals of refinery processes and how refineries fit within the wider oil industry.
- Describe the main processing units and their functions, and understand how plant configuration impacts economics and product slate.
- Recognise key crude oil characteristics and apply simple criteria for selecting appropriate feedstock for different refinery configurations.
- Calculate feedstock costs and basic refining margins at a high level, and understand what drives margins up or down.
- Understand how refining operations are optimised through planning, scheduling, unit utilisation and product blending decisions.
- Appreciate how refined products are graded and specified, and how these specifications affect blending, quality and value.
- Develop simple refinery operating and business plans that link configuration, feedstock choices, product yields and market prices.

This course is designed for technical and non-technical professionals in the oil and gas industry, business and financial institutions who want a simple, jargon-free introduction to commercial refining, including supply and trading. It is ideal for those seeking insight into how crude oil is converted into different refined products and how these products are priced and valued in open markets.

Recommended participants include:

- Oil and gas engineers
- Planning engineers and managers
- Oil marketing managers and traders
- New entrants to refinery or petroleum industry roles
- Business analysts
- Banking and government officials involved in energy or commodities
- Finance and accounting analysts supporting downstream decisions

Day 1

Introduction
- Introduction to oil industry
- Chemistry of petroleum
- Integration of refinery with oil industry

Crude Oil Characteristics
- Distillation curves
- Composition
- Fractions
- Cutting crudes
- Gravities / Sulfur content

Fundamentals of Refining Process
- Atmospheric Distillation
- Vacuum Distillation
- Catalytic Reforming
- Hydrocracking
- Cat Cracking
- Alkylation

Residue Reduction
- Thermal Cracking
- Coking
- Visbreaking
- Hydro-treating and Sulfur Plants
- Ethylene Plants

Exercises / Case Study

Day 2

Introduction to Refined Products

- ·LPG
- ·Naphtha
- ·Gasoline
- ·Jet Fuel
- ·Diesel
- ·Fuel Oil
- ·Asphalt
- ·Lubes
- ·Solvents
- ·Products grade and classification
- ·Products specification and its implication

Products Blending

- ·Gasoline blending
- ·Diesel blending
- ·Fuel oil blending
- ·Crude oil, Condensate and Natural Liquids
- ·Fuel Values – heating content
- ·Petrochemicals
- ·Derivatives and other special chemicals

Exercise / Case Study

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Day 3

Refinery Configurations

- Simple refinery
- Simple and Complex Refineries
- Impact of configuration on economics
- Refinery operating cost
- Product yields and GPW

Refinery Margins

- Gross refining margins
- Net refining margins
- Factors affecting refinery margins
- 3-2-1 crack spread
- Refinery economics and planning
- Physical markets and trading hubs
- Oil pricing and role of benchmark
- Developing refinery operating and business plans
- Project economics and cash flow

Exercise / Case Study

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